University of St. Gallen

University of St.Gallen – School of Management, Economics, Law, Social Sciences, International Affairs and Computer Science

20 Years of the Double Pukelsheim System in Switzerland: What Is the Mechanical Effect of the Transition from the Hagenbach-Bischoff Method on the Seat Distribution of Smaller Parties?

Jan Moritz Beyn

Talackerstrasse 11

8274 Tägerwilen

+41 79 551 89 39

janmoritz.beyn@student.unisg.ch

23-611-908

University of St. Gallen

Bachelor Thesis Economics

Supervisor: Dr. Aurélien Sallin

Submission date: 17.11.2026

Abstract

Table of Contents

Abstract I
Introduction 1
Theory/Literature Review 3
Hypotheses 3
    Hypothesis 1 3
    Hypothesis 2 4
Analysis 4
    Data Cleaning 5
    Results 5
        Smaller-party representation across elections 5
        Relative smaller-party representation across elections 6
        Interpretation of the Results 7
Discussion 7
Conclusion 7
Bibliography 7
    Textsources 7
    Datasources 8
Appendix 11
    Appendix 1: Explanation Data Cleaning Process 11
    Appendix 2: Code for Data Cleaning 11
    Declaration of AI-Assistance 11

Introduction

On the 18th of December 2002 the Swiss Federal Supreme Court (FSC) came to a conclusion, which disrupted the Swiss Electoral System. In BGE 129 | 185 the FSC criticized that in the elections for the Council of the City of Zurich, the thresholds for parties to win a seat, vary heavily from district to district. Through that not every voter has the same influence and therefore is the Principle of Equal Treatment (Gleichhandlungsgebot) not given anymore. Based on this criticism the court decided that the City of Zurich needs to implement corrective measures, it let the City decide which these would be. (BGE 129 I 185, E. 7.3, S. 199–200) The Government of the Canton of Zurich decided to keep the consituency structure (Regierungsrat des Kantons Zürich, 2014, pp. 3–4). Instead the canton of Zurich comissioned the german mathematician Friedrich Pukelsheim to develop a comprehensive and at the same time minimally invasive solution for the problem. Eventually Pukelsheim came up with a solution named "biproportional divisor method with standard rounding" which is today often simply called "double Pukelsheim" or "Biproportional apportionment". (Lüscher, 2024, pp. 9-10)

The main differentiation of the Pukelsheimer Proporz, is that it through the so called upper-apportionment artificially creates a single electoral district for the initial allocation of the parliamentary seats to the parties or list groups. Through this mechanism, the difference between Seat- and voteshares gets minimized and with that also the threshold for parties to enter the parliament gets lowered. (Lüscher, 2023, p. 575)

Through it's upper-apportionment mechanism and the lowered threshold, the reform is especially relevant for the representation of smaller parties. Since votes that would previously have been ineffective in small constituencies can now contribute to a seat gain at the cantonal level. Sandro Lüscher illustrates this mechanical effect with the case of Schwyz in 2016: under the old Hagenbach-Bischoff system, the Green Liberal Party would not have won a single seat, whereas under the new method they obtained three seats (Lüscher, 2025, p. 1-3).

Since the implementation through the Canton of Zurich in 2007, nine more cantons followed till date: Schaffhausen (2008), Aargau (2009), Nidwalden (2014), Zug (2014), Schwyz (2016) , Valais (2017) , Uri (2020), Graubünden (2022) in 2025 also the voters from Basel Landschaft decided to implement the Biproportional apportionment from 2027 on (Lüscher , 2024, p.17 ; Kanton Basel-Landschaft, 2025a, p.7 ; Kanton Basel-Landschaft, 2025b) While these cantons already implemented the algorithm, more cantons are currently discussing the implementation too. So is in Thurgau currently the initiative "Mehr Transparenz bei Wahlen im Thurgau" pending at the Grand Council of the Canton and special parliamentary comittee is currently examining the initiative, also in the canton of Lucerne, the iniatitive "Faire Wahlen ohne Listenflut" was submitted and the voters will be able to decide on the effectuation (Grosser Rat des Kanton Thurgaus, n.d ; Regierungsrat des Kantons Luzern, 2026, pp. 19–20). In the canton of St.Gallen in 2026 an initiative to inaugurate the algorithm was launched (Initiativkomitee «Damit jede Stimme zählt», n.d.).

The amount of cantons who have already implemented the system or are debating about doing so, shows that the decision by the FSC had an substantial impact on how Swiss Citizens elect their regional parliaments. It seems that the double-Pukelsheimer through it's wide implementation across cantons is a success story. Since the first election under double Pukelsheim

in Zurich in 2007, the system has been used in 32 cantonal parliamentary elections.[1]

Taken together, this number of elections and almost 20 years of experience provide a reason to evaluate the impact of the reform, especially on smaller parties. Has the algorithm kept its promises, and how large is the actual difference in seat allocation between the Hagenbach-Bischoff and Double Pukelsheim systems? Are there other major differences, for example in how the reform performs in smaller versus larger parliaments, and does the reform lead to a systematic increase in party-system fragmentation?

The central research gap lies in the fact that the existing literature mainly discusses the Doppelproporz in legal, normative, or system-level terms. It emphasizes improved proportionality, fewer weightless votes, and better opportunities for smaller parties, while also pointing to trade-offs such as territorial distortions and counterintuitive seat allocations (Lüscher, 2023, 2024, 2025). The closest precedents for the calculation used in this thesis are three illustrative comparisons. Pukelsheim and Schuhmacher use the actual 2011 Zurich election under Doppelproporz to calculate how the same votes would have been translated into seats under the former Hagenbach-Bischoff system, showing that larger parties receive more seats at the expense of several smaller parties under the old rule (Pukelsheim & Schuhmacher, 2011, Section D, pp. 1590–1592, Table 2). Lüscher performs the same comparison for the 2016 Schwyz election (Lüscher, 2024, p. 13, Table 3). Also Leuzinger presented in 2018 a journalistic analysis and comparison of the then seven most recently held regional elections under double Pukelsheimer, however the article in the Neue Zürcher Zeitung, doesn't provide sufficient methodological detail to reproduce choices concerning party and list-group aggregation (Leuzinger, 2018). By contrast, Lüscher's principal empirical analysis follows Swiss cantons over time and estimates how outcomes such as disproportionality and party-system fragmentation would probably have developed in the reforming cantons had they not adopted Doppelproporz (Lüscher, 2024, pp. 126-143). As discussed further in the literature review, the remaining gap is therefore the absence of a standardized application across multiple elections and cantons that measures the resulting seat gains of smaller parties.

Against this background, this thesis examines the mechanical effect of the electoral reform on the parliamentary representation of smaller parties. More specifically, it asks and shows statistically how many seats smaller parties gain through the transition from the Hagenbach-Bischoff system to Double Pukelsheim in Swiss cantonal parliamentary elections. It further examines whether the reform performs differently in smaller and larger parliaments and whether it leads to a systematic increase in party-system fragmentation. Its contribution is therefore not the invention of the counterfactual calculation, but its systematic application across 32 cantonal elections using a consistent definition of smaller parties and aggregated seat-gain measures.

[1]The 32 elections comprise Zurich (2007, 2011, 2015, 2019, and 2023; Amt für Statistik und Daten Kanton Zürich, 2026); Schaffhausen (2008, 2012, 2016, 2020, and 2024; Kanton Schaffhausen, n.d.; Staatskanzlei Kanton Schaffhausen, 2020, 2024); Aargau (2009, 2012, 2016, 2020, and 2024; Department für Finanzen und Ressourcen des Kantons Aargau, 2009; Regierungsrat des Kantons Aargau, 2013; Statistisches Amt Kanton Aargau, n.d.); Nidwalden (2014, 2018, 2022, and 2026; Staatskanzlei Nidwalden, 2014, 2018, 2022, 2026); Zug (2014, 2018, and 2022; Kanton Zug, 2022a, 2022b, 2022c); Schwyz (2016, 2020, and 2024; Bundesgericht, 2012); Valais (2017, 2021, and 2025; Kanton Wallis, 2016, 2020, 2024; Staat Wallis, 2017); Uri (2020 and 2024; Standeskanzlei Uri, 2024a, 2024b); and Graubünden (2022 and 2026; Kanton Graubünden, 2022, 2026).

Theory/Literature Review

The analytical approach most closely resembles the illustrative counterfactual comparisons conducted by Pukelsheim and Schuhmacher (2011, Section D, pp. 1590–1592, Table 2) and Lüscher (2024, p. 13, Table 3). Pukelsheim and Schuhmacher use the observed 2011 Zurich election under Doppelproporz to calculate how the same votes would have been translated into seats under the former Hagenbach-Bischoff system; their comparison shows seats moving from several smaller parties to larger parties under the old rule. Lüscher similarly compares the actual allocation in the 2016 Schwyz election with a simplified Hagenbach-Bischoff counterfactual, finding that the dominant SVP and CVP benefit under Hagenbach-Bischoff, whereas the left-wing alliance and particularly the GLP benefit from Doppelproporz. These comparisons are single-election illustrations. Lüscher's principal empirical analysis instead estimates the reform's effects on disproportionality, party-system fragmentation, turnout, and electoral competition using canton-year panel data and fixed-effects counterfactual models (Lüscher, 2024, pp. 120–143; method at pp. 127–129). The present study extends the illustrative calculation systematically across multiple cantons and elections, applies a consistent definition of smaller parties, aggregates their seat gains, and also examines fragmentation and differences by parliament size.

Hypotheses

To be able to answer the research question: 20 Years of Double Pukelsheim in Switzerland: What Is the Mechanical Effect of the Transition from Hagenbach-Bischoff on the Seat Allocation of Smaller Parties and Party-System Fragmentation?

We need to formulate hypotheses. The hypotheses are based on the theoretical considerations and the literature review. The first hypothesis focuses on the mechanical effect of the electoral reform on smaller parties, while the second hypothesis addresses the potential impact of the reform on party-system fragmentation.

Hypothesis 1

For each election e, smaller parties are those with less than 10% of the canton-wide vote. Let S_e^{DP} and S_e^{HB} be their combined seats under Double Pukelsheim and the counterfactual Hagenbach-Bischoff allocation, respectively, and let M_e be the total number of seats in that election. Their gain in seat share, shown for each election in Figure 2, is

$$\Delta_e = 100\frac{S_e^{DP} - S_e^{HB}}{M_e} \quad \text{percentage points} .$$

The main hypothesis concerns the unweighted mean of these election-level gains, $\bar{\Delta} = E^{-1}\sum_{e=1}^{E}\Delta_e$, across the E analyzed elections. Thus, "more than 5% more seats" means a mean gain greater than **five percentage points of all available seats**, rather than a 5% increase relative to the Hagenbach-Bischoff seat total.

H_0 (null hypothesis): The mean gain in smaller parties' seat share under Double Pukelsheim is at most five percentage points: $\bar{\Delta} \leq 5$.

H_1 (main hypothesis): The mean gain in smaller parties' seat share under Double Pukelsheim exceeds five percentage points: $\bar{\Delta} > 5$.

The 10% threshold is based on that the Swiss Federal Supreme Court has argued that direct and natural quorums, understood as barriers to representation within an electoral district, should generally not exceed 10% (ADD SOURCE HERE)

Hypothesis 2

$H_0$2 (null hypothesis): The transition to the double Pukelsheim system did not lead to a systematic increase in party-system fragmentation.

$$H_{0,2}: \quad \mathrm{ENP}_{\mathrm{post}} - \mathrm{ENP}_{\mathrm{pre}} \leq 0$$

$H_1$2: The transition to the double Pukelsheim system led to a systematic increase in party-system fragmentation.

$$H_{1,2}: \quad \mathrm{ENP}_{\mathrm{post}} - \mathrm{ENP}_{\mathrm{pre}} > 0$$

$$\mathrm{ENP}_v = \frac{1}{\sum_{i=1}^{n} v_i^2} \qquad \mathrm{ENP}_s = \frac{1}{\sum_{i=1}^{n} s_i^2}$$

v = votes shares, s = seat shares

Analysis

H1: The analysis takes the actual seat allocation under the Pukelsheimer Proporz as its starting point. Based on the available election data at the municipal or district level, it then reconstructs how many seats each party would have obtained under the former Hagenbach-Bischoff system. The comparison between the actual seat distribution under the new system and the hypothetical distribution under the old system makes it possible to identify how many seats each party gained or lost as a result of the reform. In this way, the study assesses the mechanical effect of the electoral-system change on party representation, with a particular focus on smaller parties.

For the purposes of this thesis, a smaller party is defined as an electoral list or list group with less than 10% of the canton-wide vote share in the relevant election. Youth lists and other special-interest lists, such as women's lists or pensioners' lists, are treated as separate analytical units whenever they are reported separately in the official election results; their votes and seats are not attributed to a mother party. Accordingly, the term *party* in the analysis refers to the standardized electoral list or list group recorded in the dataset and does not necessarily correspond to a broader organizational party family. The choice of 10% as the threshold is grounded in both legal and normative considerations. In its de-cision BGE 131 I 74, the Swiss Federal Supreme Court argued that direct and natural quorums, understood as barriers to representation within an electoral district, should generally not exceed 10%. The 2013 report of the Federal Chancellery refers to the same benchmark and further notes that the Venice Commission considers thresholds in the range of 10–15% to be the upper acceptable limit . Using vote share rather than absolute party size is particularly appropriate because a party may be electorally small in one region while being comparatively strong in another. Low thresholds are closely linked to the principle of Erfolgswertgleichheit, or equality of vote success value, which requires that votes should be translated into seats as equally and effectively as possible. Investigating the threshold is especially important, since none of the reform cantons had a <10% threshold in all districts before the implantation of the new electoral reform (Lüscher, 2024, p. 115).

Data Cleaning

To be able to analyze the data from the different elections and compare the outcomes. each dataset has to be cleaned. The data For some cantons e.g. Valais and Schwyz the data is very scattered, therefore for there multiple datasets per election have to be cleaned and merged.

The procedure for the data cleaning was the following: For each canton in alphabetically order, for each election the data was cleaned.

To ensure the compatibility of the different datasets, the column names were standardized. The column names were renamed for each dataset to the following: “total_votes”, “svp_votes”, “sp_votes”, “cvp_votes”, “fdp_votes”, “glp_votes”, “bdp_votes”, “green_votes”, “other_votes”. It might be that some of the datasets do not contain all of these columns, since the party landscapes varies geographically and over time.

For the counterfactual, the district-level party votes and district seat magnitudes from the pre-Pukelsheim era are retained. The Hagenbach-Bischoff allocation follows the relevant cantonal law of the time (e.g., Sections 12-13 of the Aargau Grossratswahlgesetz for the 2005 election). The main counterfactual treats every party independently, ignoring list alliances, as they were abolished with the transition to Double Pukelsheim. Vote shares use each canton’s *Wählerzahl* definition: district-level party votes are divided by the district seat magnitude and rounded to the nearest integer before cantonal aggregation.

The official Double Pukelsheim results (`seats_dp`) are sourced from official cantonal authorities and are documented in the Bibliography. A single lookup table consolidates all official results for easy merging with the Hagenbach-Bischoff counterfactuals. All data processing is implemented in canton-specific R scripts (R/canton_*.R) which apply shared functions for algorithm execution and validation.

Results

Smaller-party representation across elections

Figure 1 compares the observed Double Pukelsheim allocation with the counterfactual Hagenbach-Bischoff allocation for every cleaned election. For each election, the figure aggregates the seats and canton-wide vote shares of all parties below the 10% threshold defined above. The grey line connects the two allocations derived from the same election.

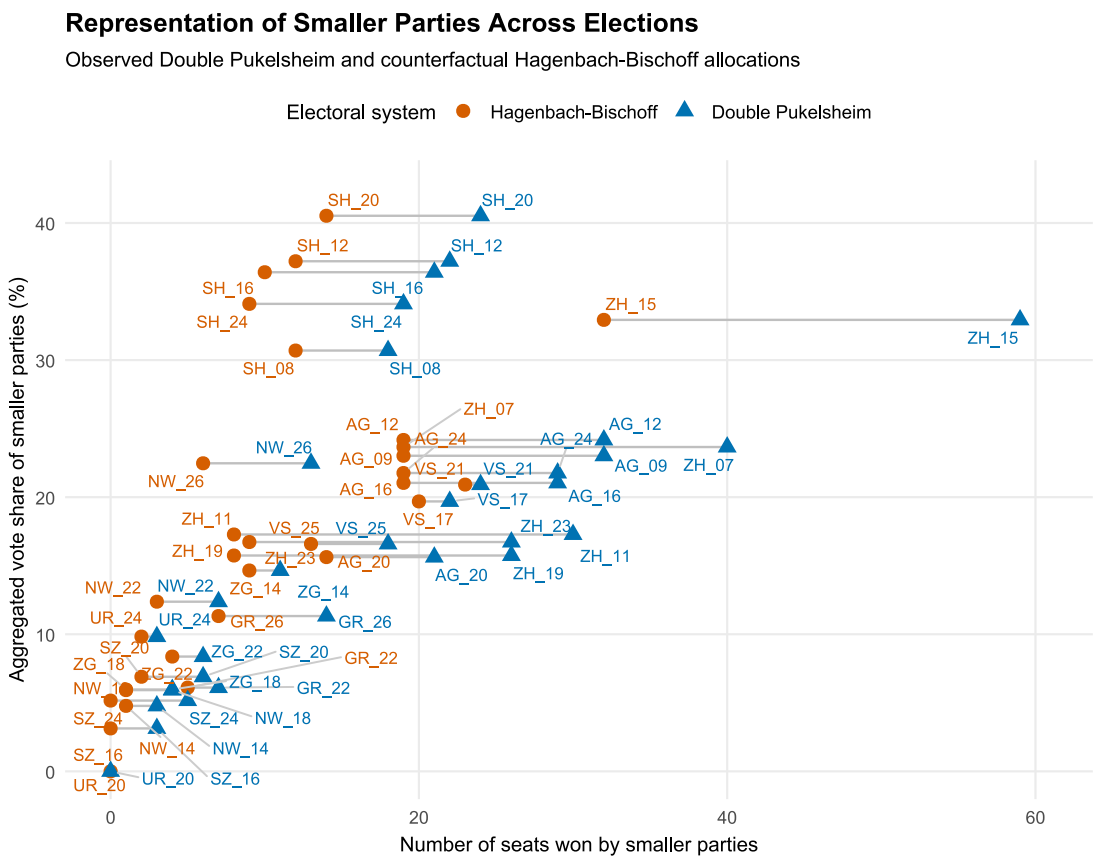


Figure 1: Seats won by smaller parties under Double Pukelsheim and the counterfactual Hagenbach-Bischoff allocation. Smaller parties are those with less than 10% of the canton-wide vote share.

Relative smaller-party representation across elections

Figure 2 shows the same elections using the percentage of all parliamentary seats won by smaller parties on the x-axis. Each label reports the gain under Double Pukelsheim as a percentage-point share of all available seats.

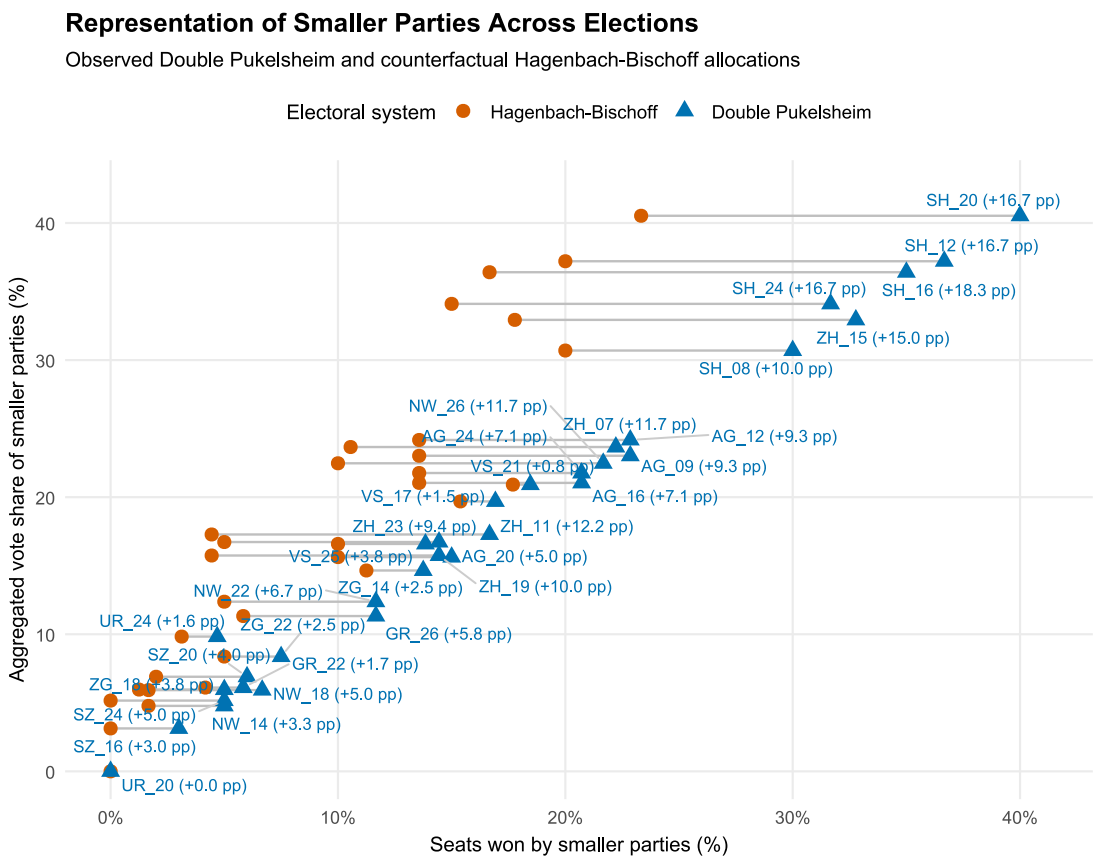


Figure 2: Percentage of seats won by smaller parties under Double Pukelsheim and counterfactual Hagenbach-Bischoff.

> **i** Hypothesis 1 summary
>
> Across all 32 elections, the unweighted mean gain in smaller parties' share of all seats was 7.4 percentage points (Q1: 3.2; median: 6.2; Q3: 10.4). The mean weighted by each election's parliament size was 7.5 percentage points. 17 individual elections exceeded five percentage points.

Interpretation of the Results

Discussion

Conclusion

Bibliography

Textsources

Bundesgericht. (2002). *BGE 129 I 185*. https://relevancy.bger.ch/php/clir/http/index.php?highlight_docid=atf%3A%2F%2F129-I-185%3Ade&lang=de&type=show_document

Grosser Rat des Kanton Thurgaus. (n.d.). *Pendente Geschäfte – Elektronische Geschäftsdatenbank GRGEKO*. Retrieved September 15, 2026, from https://grgeko.tg.ch/web/grgeko/pendente-geschaefte

Initiativkomitee "Damit jede Stimme zählt." (n.d.). *Initiative für ein faires Wahlsystem im Kanton St. Gallen*. Retrieved September 15, 2026, from https://jede-stimme-zaehlt.ch/

Kanton Basel-Landschaft. (2025a). *Abstimmung vom 9. Februar 2025: Abstimmungsvorlagen [Abstimmungsbroschüre]*. https://abstimmungen.bl.ch/app/archive/de/storage/cd604418d5ac943f3373dd673738f3101f41769264c57eba9d692ea3d9a12347.pdf

Kanton Basel-Landschaft. (2025b). *Einführung des kantonalen Doppelproporzes im Kanton Basel-Landschaft – Teilrevision des Gesetzes über die politischen Rechte [Abstimmungsresultat vom 9. Februar 2025]*. https://abstimmungen.bl.ch/app/archive/de/vote/ct-13-100.html

Leuzinger, L. (2018, September 10). *Wie das Doppelproporz-Wahlsystem die Chancen der kleineren Parteien erhöht*. Neue Zürcher Zeitung. https://www.nzz.ch/schweiz/konkurrenz-fuer-die-grossen-parteien-ld.1409175

Lüscher, S. (2023). *Parlamentswahlen nach dem doppelt-proportionalen Sitzzuteilungsverfahren im Lichte der territorialen Repräsentation: Eine empirische Annäherung an ein konzeptionelles Dilemma*. Zeitschrift für Parlamentsfragen, 54(3), 571–595. https://doi.org/10.5771/0340-1758-2023-3-571

Lüscher, S. (2024). *Redeeming the pledge of electoral equity? A study of the roads to and effects of the divisor-based biproportional apportionment method in Switzerland* [Doctoral dissertation, University of Zurich]. https://doi.org/10.5167/uzh-264856

Lüscher, S. (2025, February 20). "20 Jahre Doppelproporz—eine Rückschau". DeFacto. https://www.zora.uzh.ch/entities/publication/9b83ccf9-a61f-45ff-b0c8-c52b5cfcb1c7

Pukelsheim, F., & Schuhmacher, C. (2011). *Doppelproporz bei Parlamentswahlen – ein Rück- und Ausblick*. Aktuelle Juristische Praxis / Pratique Juridique Actuelle, 20(12), 1581–1599. https://www.math.uni-augsburg.de/htdocs/emeriti/pukelsheim/2011e.pdf

Regierungsrat des Kantons Luzern. (2026, March 31). Volksinitiativen «Faire Wahlen ohne Listenflut» und «Stopp Listenflut»: Botschaft des Regierungsrates an den Kantonsrat (B 93). https://www.lu.ch/-/klu/ris/cdws/document?fileid=641a78c026024e88bd438d2f0d3d0462

Regierungsrat des Kantons Zürich. (2014, November 26). *Postulat (Änderung Wahlgesetz [Wahlkreiseinteilung]) (RRB Nr. 1254/2014; KR-Nr. 219/2014)*. https://parlzhcdws.cmicloud.ch/parlzh5/cdws//Files/89dffabbc7a240f08e824f7a30bebf4d-332/1/pdf

Datasources

Amt für Statistik und Daten Kanton Zürich. (2026). *Archiv Wahlresultate Kantonsratswahlen [Data set]*. Kanton Zürich. https://www.zh.ch/de/politik-staat/wahlen-abstimmungen/wahlarchiv.zhweb-noredirect.zhweb-cache.html#/datasets/690@statistisches-amt-kanton-zuerich

Bundesgericht. (1965). *BGE 91 I 266*. https://relevancy.bger.ch/php/clir/http/index.php?highlight_docid=atf%3A%2F%2F91-I-266%3Ade&lang=de&type=show_document

Bundesgericht. (2010). *BGE 136 I 352*. https://relevancy.bger.ch/php/clir/http/index.php?highlight_docid=atf%3A%2F%2F136-I-352%3Ade&lang=de&type=show_document

Bundesgericht. (2012, 19. März). *Urteil 1C_407/2011, 1C_445/2011 und 1C_447/2011: Kantonsratswahlen 2012, Verletzung des Wahlrechts*. https://search.bger.ch/ext/eurospider/live/de/php/aza/http/index.php?highlight_docid=aza%3A%2F%2F19-03-2012-1C_407-2011&lang=de&type=show_document

Department für Finanzen und Ressourcen des Kantons Aargau. (2009, April 21). *Detailergebnisse der Grossratswahlen vom 8. März 2009*. https://www.ag.ch/de/medien/medienmitteilungen?mm=detailergebnisse-der-grossratswahlen-vom-8-maerz-2009-742f7d21-d4b0-454f-abd7-25471ef7f566_de

Kanton Aargau. (1988/2007). *Gesetz über die Wahl des Grossen Rates (Grossratswahlgesetz; GRWG), § 8 Abs. 1*. https://gesetzessammlungen.ag.ch/api/de/versions/3570/pdf_file

Kanton Aargau. (2004). *Grossratswahlgesetz: Änderung vom 8. Juni 2004*. Abstimmungsbroschüre zur kantonalen Volksabstimmung vom 26. September 2004, pp. 5–8. https://www.ag.ch/media/kanton-aargau/portal/aktuell/wahlen-abstimmungen/abstimmungen/abstimmungsbroschueren/20040926-abstimmungsbroschuere.pdf

Kanton Aargau. (2008). *Gesetz über die Wahl des Grossen Rates (Grossratswahlgesetz), Fassung gemäss Gesetz vom 18. September 2007, in Kraft seit 1. Juli 2008*. https://gesetzessammlungen.ag.ch/api/de/versions/1234/pdf_file

Kanton Graubünden. (2022). *Pressedaten Proporzwahl kantonal: Grossratswahlen 2022* [Excel spreadsheet]. https://www.gr.ch/DE/publikationen/abstimmungenwahlen/Grossratswahlen-2022/ResultateGrossratswahlen/20220515_Pressedaten_Proporzwahl_kantonal.xlsx

Kanton Graubünden. (2026). *Pressedaten Proporzwahl kantonal: Grossratswahlen 2026* [Excel spreadsheet]. https://www.gr.ch/DE/publikationen/abstimmungenwahlen/grw26/Resultate1/20260614_Pressedaten_Proporzwahl_kantonal.xlsx

Kanton Schaffhausen. (n.d.). *Frühere Urnengänge (seit 2003)*. https://archiv.sh.ch/daten/Fruehere-Urnengaenge-seit-200.1127.0.html

Kanton Uri. (2018). *Synoptische Darstellung zum Gesetz über die Verhältniswahl des Landrats (Proporzgesetz)*. https://www.ur.ch/_docn/113506/LA.2017-0413_IV._Synoptische_Darstellung_Proporzgesetz.pdf

Kanton Wallis. (2004a). *Gesetz über die politischen Rechte (kGPR) vom 13. Mai 2004* (SGS 160.1; Stand 9. April 2017). https://lex.vs.ch/app/de/texts_of_law/160.1/versions/1375

Kanton Wallis. (2016, 29. September). Legislaturperiode 2017–2021: Verteilung der Anzahl Grossratssitze pro Bezirk. https://www.vs.ch/documents/529400/2263784/MM_Verteilung_Grossratssitze.pdf/30be6442-6e9e-4d91-adef-76d3f60d6107

Kanton Wallis. (2020, 25. September). Legislaturperiode 2021–2025: Verteilung der Anzahl Grossratssitze pro Bezirk. https://www.vs.ch/documents/529400/8932772/2020+09+25+-+MM+-+Verteilung+Grossratssitze.pdf/62d59d10-2075-5231-e385-88688c7cb559

Kanton Wallis. (2024, 20. September). Anhang zur Verteilung der Anzahl Grossratssitze pro Bezirk, Legislaturperiode 2025–2029. https://www.vs.ch/documents/529400/33671442/Anhang+Verteilung.pdf/b1f92f33-f4b2-35ef-6149-838af6c70b09

Kanton Zürich. (1983/2005). *Gesetz über die Wahlen und Abstimmungen (Wahlgesetz)* (LS 161; Nachtrag 47, Stand 1. Januar 2005), §§ 87–88b. https://www.notes.zh.ch/appl/zhlex_r.nsf/WebView/C4AB55033822301DC1256F4A0047A916/$File/161_4.9.83_47.pdf

Kanton Zug. (2006, September 28). *Gesetz über die Wahlen und Abstimmungen (Wahl- und Abstimmungsgesetz, WAG)* (Vorlage Nr. 1300.24, Laufnummer 12 198). https://kr-geschaefte.zug.ch/dokumente/2793/pdoc_1494_1.pdf

Kanton Zug. (2022a, July 1). *Kantonsratswahl 2018 [Data set]*. https://wab.zug.ch/elections/kantonsratswahl-2018/data

Kanton Zug. (2022b, July 1). *Kantonsratswahl Kanton Zug [Data set]*. https://wab.zug.ch/elections/kantonsratswahl-kanton-zug-2014/data

Kanton Zug. (2022c, October 2). *Kantonsratswahl 2022 [Data set]*. https://wab.zug.ch/elections/kantonsratswahl-2022/data

Regierungsrat des Kantons Aargau. (2013). *Jahresbericht mit Jahresrechnung 2012. Kanton Aargau.* https://www.ag.ch/media/kanton-aargau/dfr/dokumente/finanzen/planung-und-berichterstattung/jahresbericht-mit-jahresrechnung/botschaft-2012.pdf

Regierungsrat des Kantons Schaffhausen. (2007, April 24). *Bericht und Antrag des Regierungsrates des Kantons Schaffhausen an den Kantonsrat betreffend Anpassung des Wahlsystems für den Kantonsrat (Bericht und Antrag 07-41).* https://archiv.sh.ch/daten/fileadmin/Redaktoren/Dokumente_nicht_im_Formularpool/Regierung/Vorlagen/2007/24-4-07.pdf

Staat Wallis. (2017). *Grossräte 2017 – Resultate: Offizielle Endresultate. Archivierte Abstimmungs- und Wahlresultate.* https://vework-public.vs.ch/app/archive/sites/votel2/DE/1/result/gc/137

Staatskanzlei des Kantons Aargau. (2008, September 15). *Grossratswahlen: Vorverfahren gestartet.* https://www.ag.ch/de/medien/medienmitteilungen?mm=Grossratswahlen-Vorverfahren-gestartet-50a84551-3d6d-454c-a33c-5644c8bf50a5_de

Staatskanzlei des Kantons Aargau. (2012, April 5). *Gemeinsame Grossrats- und Regierungsratswahlen: Vorverfahren gestartet: Anmeldefristen sind bekannt.* https://www.ag.ch/de/medien/medienmitteilungen?mm=gemeinsame-grossrats-und-regierungsratswahlen-e7d48210-c7f0-46ac-8a70-ffc1aab7b4d6_de

Staatskanzlei des Kantons Aargau. (2016, March 18). *Ausschreibung der Grossrats- und Regierungsratswahlen 2016: Unterlagen unter www.ag.ch/wabag verfügbar.* https://www.ag.ch/de/medien/medienmitteilungen?mm=ausschreibung-der-grossrats-und-regierungsratswahlen-2016-f7404346-a5eb-46e0-8627-301716c82855_de

Staatskanzlei des Kantons Aargau. (2005, March 7). *Bezirksprotokolle über die Erneuerungswahl des Grossen Rats für die Amtsperiode 2005/2009: Wahltag: 27. Februar 2005 (Amtsblattbeilage 2005-1).* https://doczz.net/doc/5954970/amtsblattbeilage_vom_07.03.2005

Staatskanzlei Nidwalden. (2014). *Wahlergebnis der Landratswahlen vom 23. März 2014 [Unpublished raw data].* Kanton Nidwalden.

Staatskanzlei Nidwalden. (2018). *Wahlergebnis der Landratswahlen vom 4. März 2018 [Unpublished raw data].* Kanton Nidwalden.

Staatskanzlei Nidwalden. (2022). *Wahlergebnis der Landratswahlen vom 13. März 2022 [Unpublished raw data]*. Kanton Nidwalden.

Staatskanzlei Nidwalden. (2026). *Wahlergebnis der Landratswahlen vom 8. März 2026 [Unpublished raw data]*. Kanton Nidwalden.

Staatskanzlei Kanton Schaffhausen. (2020, September 27). *Resultate der Kantonsratswahl vom 27. September 2020*. Kanton Schaffhausen. https://sh.ch/CMS/Webseite/Kanton-Schaffhausen/Beh-rde/Abstimmungen-und-Wahlen-6275841-DE.html

Staatskanzlei Kanton Schaffhausen. (2024, September 22). *Kantonsratswahlen vom 22. September 2024 – Resultate*. Kanton Schaffhausen. https://www.sh.ch/CMS/Webseite/Kanton-Schaffhausen/Beh-rde/Regierung/Staatskanzlei-15401466-DE.html

Standeskanzlei Uri. (2024a). *AuswertungKapitel2-2024* [Unpublished Excel Workbook]. Kanton Uri.

Standeskanzlei Uri. (2024b). *Landratswahlen 2024: Statistische Auswertung*. Kanton Uri. https://www.ur.ch/_docn/384232/Landratswahlen_2024_stat_Auswertung.pdf

Statistisches Amt Kanton Aargau. (n.d.). *Grossratswahlen: Detailresultate, ab 1985 [Data set]*. Kanton Aargau. Retrieved September 1, 2026, from https://www.ag.ch/app/staag/statistikportal/

Appendix

Appendix 1: Explanation Data Cleaning Process

Appendix 2: Code for Data Cleaning

Declaration of AI-Assistance